

The empirical recurrence for OEIS A232035, recovered from its tail

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Abstract

OEIS A232035 records “Empirical recurrence of order 95” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 155$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 3$, so the recurrence is proved for every $n \geq 98$. Its last coefficient is $c_{95} = 16 \neq 0$, so no further tail satisfies anything shorter; any order-95 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A232035 is “Number of $(n+1) \times (5+1)$ 0..1 arrays with every element equal to some horizontal, vertical or antidiagonal neighbor, with top left element zero.”. It has offset 1 and begins 450, 24719, 1272579, 66684606, 3495632937, 183196162778, 9600712297741, 503143350126954, ...

The entry states:

Empirical recurrence of order 95 (see link above)

As of the “Last modified” line on the live entry (Jul 23 2025 06:49:19, revision 7) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 155$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 155$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 95: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 3$ is the first whose minimal order is exactly the stated 95.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 310$ exact terms from $n = 3$ on, it returns order 95 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{95} c_i a(n-i),$$

c_1	64	c_{25}	−41258568430120824	c_{49}	−1687871305857205955	c_{73}	10127923234905
c_2	−580	c_{26}	−69138949616588022	c_{50}	−1486795945890588081	c_{74}	−4429389024493
c_3	−1831	c_{27}	152488105606473582	c_{51}	458380610254097787	c_{75}	2042163544563
c_4	18240	c_{28}	293430883725310268	c_{52}	11741612855504503	c_{76}	−793107234613
c_5	103788	c_{29}	−483935935518017843	c_{53}	1842968336936970161	c_{77}	379479678977
c_6	241498	c_{30}	−894305650751685475	c_{54}	−1141034412676844611	c_{78}	−152434138282
c_7	−5738648	c_{31}	1274637474826307315	c_{55}	1191032637809714817	c_{79}	59056263739
c_8	−27723988	c_{32}	1995251894340382093	c_{56}	−1376143708257249263	c_{80}	−25430239283
c_9	130842842	c_{33}	−2593972484508465470	c_{57}	814592113468500780	c_{81}	9116872665
c_{10}	895171032	c_{34}	−3249721161070025185	c_{58}	−666254160467550319	c_{82}	−3455928217
c_{11}	−67732285	c_{35}	3683865866724839046	c_{59}	445972495890725481	c_{83}	1203767834
c_{12}	−18064353051	c_{36}	3816939442647213540	c_{60}	−233007068061067517	c_{84}	−398147065
c_{13}	−65244483298	c_{37}	−2827538245923567843	c_{61}	147408587686722799	c_{85}	140570409
c_{14}	255401207973	c_{38}	−3266100828549812342	c_{62}	−69953804537762106	c_{86}	−39787754
c_{15}	1749925851443	c_{39}	−749634969155498588	c_{63}	33015615908831194	c_{87}	12130914
c_{16}	−2498514848646	c_{40}	1987588829882974350	c_{64}	−16777534183165444	c_{88}	−3352233
c_{17}	−25627445412157	c_{41}	4428658118980814458	c_{65}	6440598676123553	c_{89}	870557
c_{18}	14110982708916	c_{42}	−250964438196289812	c_{66}	−3220034974075906	c_{90}	−228956
c_{19}	249983820789268	c_{43}	−4124831740673375867	c_{67}	1498607534351212	c_{91}	40796
c_{20}	9549321883813	c_{44}	−2098538777948093565	c_{68}	−569560364034057	c_{92}	−8060
c_{21}	−1759225070990253	c_{45}	478214080015989862	c_{69}	319583643301660	c_{93}	752
c_{22}	−1054826661165387	c_{46}	2769399816182308862	c_{70}	−130576351727809	c_{94}	−48
c_{23}	9470046266448039	c_{47}	2310901712455592177	c_{71}	50086030281705	c_{95}	16
c_{24}	11144158069568714	c_{48}	−843902917793308692	c_{72}	−26699615112261		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 98$, and it is the recurrence OEIS A232035 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 3$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{95} - c_1x^{95-1} - \dots - c_{95}$. Its constant term is $-c_{95} = -16$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k\lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 95 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 95, so $\deg q = 95$, and it is monic. It holds from some index t on. From $\max(t, 3)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 95, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 13 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 95, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is 16.

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-95 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

- [1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, <https://oeis.org>, 2026. Sequence A232035.
- [2] J. L. Massey, Shift-register synthesis and BCH decoding, *IEEE Trans. Inform. Theory* **15** (1969), 122–127.
- [3] R. P. Stanley, *Enumerative Combinatorics*, Volume 1, 2nd ed., Cambridge University Press, 2011. (Transfer-matrix method, Section 4.7.)
- [4] M. Kauers and P. Paule, *The Concrete Tetrahedron*, Springer, 2011. (C-finite sequences and their closure properties.)
- [5] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.